

MODULE 5

GLOBAL WARMING & EIA

Conservation of Natural Resources

Course Code: BCV755B

Terra Study Materials
Department of Civil Engineering



Complete Study Notes

Global Warming • Greenhouse Effect • Indicators & Causes • Effects • Global Climate Change •
Health Impacts • Environmental Impact Assessment • EIA Cycle • EIA Components

0. Contents

1	Concept of Global Warming	3
1.1	The Journey of Sunlight (Step by Step)	3
1.2	When Does the Problem Begin?	3
2	Indicators of Global Warming	4
3	Factors Affecting Global Warming (Causes)	5
3.1	Factor 1 – Carbon Dioxide (CO ₂)	5
3.2	Factor 2 – Methane (CH ₄)	6
3.3	Factor 3 – Nitrous Oxide (N ₂ O)	6
3.4	Factor 4 – Deforestation	6
4	Effects of Global Warming	7
5	What is Global Climate Change?	8
6	Indicators of Climate Change	9
6.1	Global Warming	9
6.2	Changes in Polar and Glacial Ice	9
6.3	Ocean Acidity	9
6.4	Climate and Health	9
6.5	Changing Wind Patterns	10
6.6	Changing Precipitation Patterns	10
6.7	Storm Intensity and Frequency	10
6.8	Changing Biomes	10
7	Effect of Global Climate Change on Human Health	10
8	Introduction to EIA	11
8.1	What EIA Does	11
8.2	Purpose of EIA	12
9	Three Core Values of EIA	12
10	Benefits of Environmental Assessment	12
11	Evolution and History of EIA	12
12	EIA – Cycle and Procedure in India	13
12.1	Phase 1 – Screening	13
12.2	Phase 2 – Scoping (and Alternatives)	13
12.3	Phase 3 – Baseline Data Collection	14
12.4	Phase 4 – Impact Analysis	14
12.5	Phase 5 – Alternatives, Mitigation & EIA Report	14

12.5.1 Assessment of Alternatives	14
12.5.2 Delineation of Mitigation Measures	15
12.5.3 EIA Report	15
12.6 Phase 6 – Public Hearing	15
12.7 Phase 7 – Environmental Management Plan (EMP)	15
12.8 Phase 8 – Decision Making	15
12.9 Phase 9 – Monitoring the Clearance Conditions	16
13 EIA Process – Flowchart	16
14 Components of EIA (EIA Report Contents)	16
14.1 Comprehensive EIA vs. Rapid EIA	16
14.2 A. Air Environment	17
14.3 B. Noise Environment	17
14.4 C. Water Environment	17
14.5 D. Biological Environment	18
14.6 E. Land Environment	18
14.7 F. Socio-economic and Health Environment	18
14.8 G. Risk Assessment	18
14.9 H. Environmental Management Plan (EMP)	19
15 EIA – Benefits and Flaws	19
15.1 Benefits of EIA	20
15.2 Flaws of EIA	20
Summary Table	20

1. Concept of Global Warming

Global Warming

Global warming is the **gradual increase in the average temperature** of the Earth's atmosphere and oceans.

To understand this properly, let us trace the complete **journey of sunlight** on Earth.

1.1 The Journey of Sunlight (Step by Step)

1. **Sunlight reaches Earth:** when solar radiation reaches the Earth, not all of it is absorbed.
2. **Reflection:** about **30% of sunlight** is sent back into space by clouds, atmospheric particles, reflective ground surfaces (snow, ice) and the surface of oceans.
3. **Absorption:** the remaining **70%** is absorbed by the oceans, the air (atmosphere) and land surfaces. This absorbed energy **heats the planet's surface and atmosphere**, making life possible.
4. **Re-radiation:** as the Earth warms, it radiates this energy back to space as **thermal radiation and infrared rays** – a process that naturally **cools the Earth**.
5. **The (natural) Greenhouse Effect:** some of this outgoing radiation is **re-absorbed** by certain gases – carbon dioxide (CO₂), water vapour (H₂O), ozone (O₃), methane (CH₄) and other trace gases. They **re-radiate the heat back** to the surface. These are called **Greenhouse Gases (GHGs)** because of their heat-trapping capacity – just like a greenhouse traps heat to keep plants warm.

The Natural Greenhouse Effect is GOOD

The natural greenhouse effect is actually **essential for life**. Without greenhouse gases, the Earth's average surface temperature would be around **–18°C** instead of the current **+15°C** – life as we know it would not be possible.

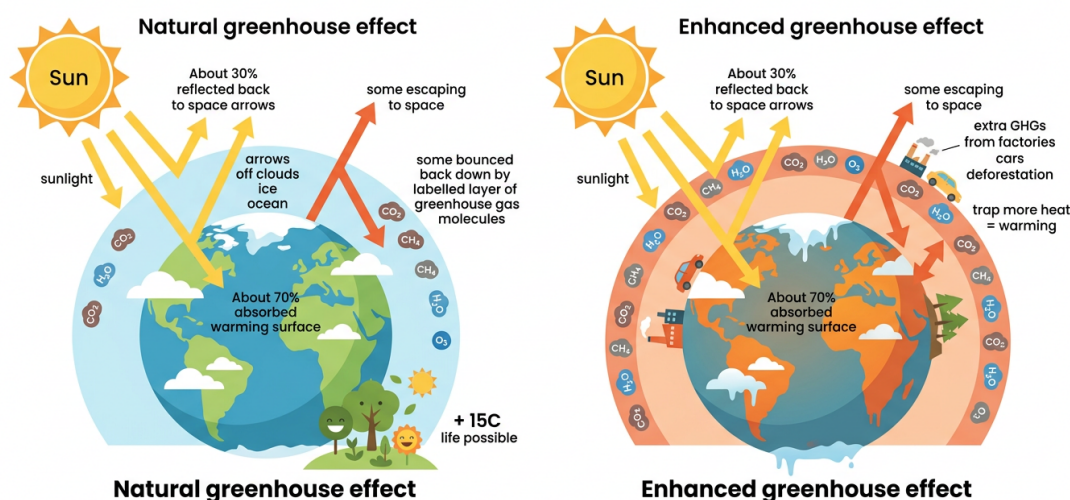
1.2 When Does the Problem Begin?

- The problem started when **human activities artificially increased** the concentration of GHGs at an alarming rate over the past two centuries (since the Industrial Revolution).
- As of **2004**, over **8 billion tons of CO₂** were being pumped into the atmosphere annually.
- This increased GHG concentration **traps more outgoing thermal radiation**, preventing it from escaping into space.
- This leads to the “**Human-Enhanced Global Warming Effect**” or “**Enhanced Greenhouse Effect**.”

Conclusion

Scientific observations confirm that it is the **human-enhanced greenhouse effect** that is causing the planet to heat up. The planet has experienced the **largest increase in surface temperature over the last 100 years**.

THE GREENHOUSE EFFECT



The greenhouse effect – natural versus human-enhanced.

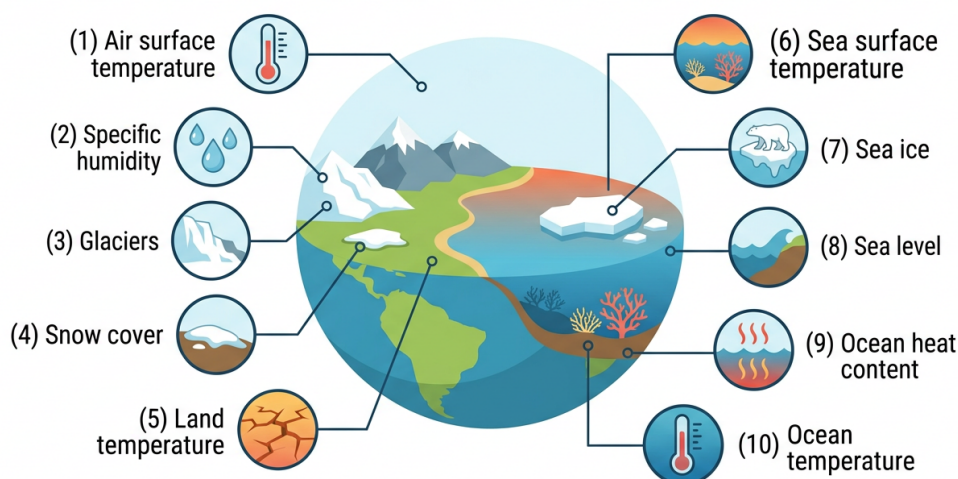
2. Indicators of Global Warming

Indicators of global warming are **measurable signs** that clearly show the Earth is warming. Ten indicators are recognised.

1. **Air Surface Temperature:** the most direct indicator; global average air temperatures have risen steadily, with a consistent upward trend over the past century. Even a rise of 1–2°C has profound consequences.
2. **Specific Humidity:** the amount of water vapour in air. As temperatures rise, more water evaporates, increasing water vapour. Since water vapour is itself a GHG, this creates a **positive feedback loop** – more warming → more evaporation → even more warming.
3. **Glaciers:** glaciers worldwide (Himalayas, Alps, Andes, Arctic, Antarctic) are **melting and retreating** at unprecedented, accelerating rates – raising sea levels and threatening freshwater supplies. One of the most visible, dramatic indicators.
4. **Snow Cover:** the extent and duration of Northern-Hemisphere snow cover is decreasing (melting earlier, arriving later). Less snow means less sunlight reflected (reduced albedo) – another positive feedback loop.
5. **Temperature over Land:** land warms **faster than oceans** (lower heat capacity). The Arctic warms at about **twice the global average rate**.
6. **Sea Surface Temperature (SST):** oceans absorb much heat; SST has risen steadily, fuelling stronger storms, causing **coral bleaching**, and linking to El Niño / La Niña patterns.
7. **Sea Ice:** Arctic sea ice extent and thickness are declining dramatically (about **13% per decade**). Exposed dark ocean absorbs more heat; threatens polar bears and seals.
8. **Sea Level:** rising due to (a) **thermal expansion** of warming water and (b) **melting glaciers and ice caps** – threatening coastal cities and low-lying islands.

9. **Ocean Heat Content:** oceans act as a heat sink, absorbing over **90% of the excess heat**. Rising ocean heat content is one of the most definitive indicators that Earth is gaining energy.
10. **Temperature over Oceans:** ocean-surface temperatures are rising too, affecting evaporation, precipitation and marine life, and influencing monsoon systems critical to India.

Indicators of Global Warming



The ten indicators of global warming.

3. Factors Affecting Global Warming (Causes)

The primary cause of global warming is the **emission of greenhouse gases** by human activities.

Share of GHG Emissions

Carbon dioxide (CO₂) contributes about **80%**; **Methane (CH₄)** and **Nitrous Oxide (N₂O)** together contribute the remaining **20%**.

3.1 Factor 1 – Carbon Dioxide (CO₂)

Sources:

- **Power Plants:** burning fossil fuels (coal, oil, gas) for electricity is the **single largest source**; power stations contribute about **21.3%** of annual GHG emissions.
- **Transportation:** about **20% of CO₂** comes from petrol/diesel engines; transport fuels account for about **14%** of total emissions.
- **Industrial Processes:** combustion and chemical reactions contribute about **16.8%**.
- **Deforestation:** cutting and burning forests releases stored CO₂ and destroys the carbon sink.

Why CO₂ is so significant

CO₂ stays in the atmosphere for **hundreds to thousands of years**. Its concentration has risen from about **280 ppm** (pre-industrial) to over **420 ppm** today – the highest in 800,000 years.

3.2 Factor 2 – Methane (CH₄)**Potency**

Methane is **more than 20 times more effective than CO₂** at trapping heat (over 100 years) – a very potent GHG even in small quantities.

Sources:

- **Rice Paddies:** anaerobic decomposition in flooded fields.
- **Bovine Flatulence (Livestock):** enteric fermentation in cattle, sheep and other ruminants; agricultural byproducts contribute about **12.5%**.
- **Bacteria in Bogs (Wetlands):** natural decomposition without oxygen.
- **Fossil-Fuel Manufacture & Distribution:** leaks from coal mines, gas pipelines and oil drilling; fossil-fuel retrieval accounts for about **11.3%**.
- **Landfills / Waste Disposal:** decomposing organic waste; waste disposal and treatment contributes about **3.4%**.

3.3 Factor 3 – Nitrous Oxide (N₂O)**Potency**

Nitrous oxide is about **298 times more powerful** than CO₂ (100-year global warming potential) and remains in the atmosphere for about **114 years**.

Sources:

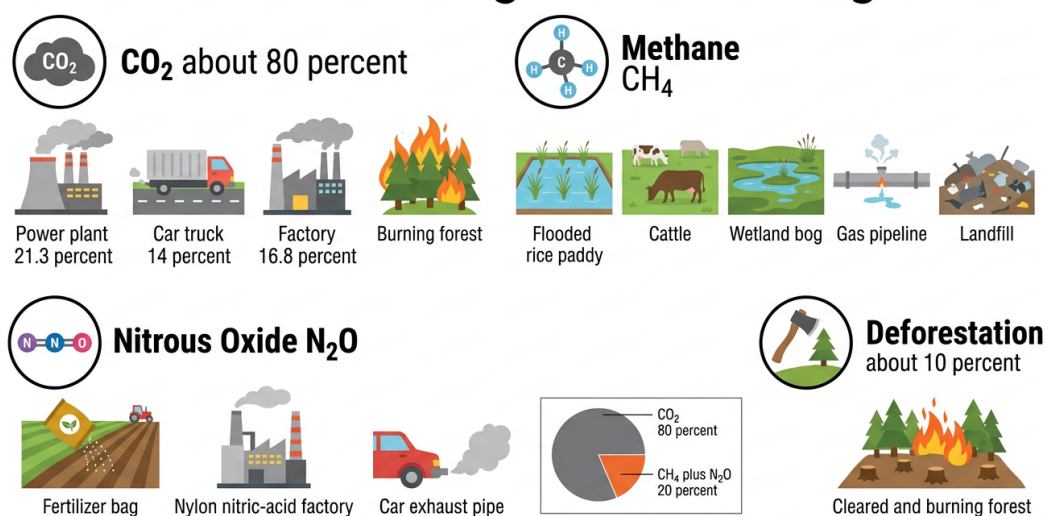
- **Agricultural Fertilizers:** nitrogen fertilizers release N₂O from soils via microbial processes – the **largest source**.
- **Nylon & Nitric Acid Production:** released as an industrial byproduct.
- **Cars with Catalytic Converters:** some converters produce N₂O as a byproduct.
- **Burning of Organic Matter:** burning agricultural waste, biomass and forests.

3.4 Factor 4 – Deforestation

- Forests are cut and burned for residential use (urban expansion), industrialisation and agriculture.
- **Direct emission of CO₂:** burning releases stored carbon.
- **Loss of carbon sinks:** fewer trees to absorb CO₂.
- **Albedo changes:** dark forest soil absorbs more heat than the reflective canopy.

- Land use and biomass burning contribute about **10%** of annual GHG emissions.

Factors Causing Global Warming



Sources of greenhouse-gas emissions.

4. Effects of Global Warming

Global warming has wide-ranging, serious consequences:

- Frequent Temperature Extremes (Killer Heat Waves):** more frequent, intense and longer heat events causing heat strokes, dehydration and death – especially among the elderly, children and outdoor workers.
- Changing Rainfall Patterns:** the water cycle is altered – some regions get more rain, others severe drought; disrupting agriculture, water supply and ecosystems.
- Rise in Sea Levels:** melting glaciers and thermal expansion put low-lying coasts, islands and deltas at risk of permanent inundation.
- Frequent Storms and Coastal Flooding:** warmer oceans give more energy to tropical storms, hurricanes and cyclones.
- Changes in Regional Climate:** climate zones shift – forests die or migrate, crop yields decline, glacier-fed rivers may shrink.
- Drought:** prolonged and more severe, especially in subtropical areas, causing water scarcity.
- Food Shortages:** shifting agricultural zones plus population growth lead to food insecurity and famine.
- Greater Warming near the Poles:** the Arctic and Antarctic warm at **twice the global average rate** – called **polar amplification**.
- Air Pollution Made Worse:** higher temperatures increase ground-level ozone (smog) and particulate matter.

10. **Respiratory Complications:** asthma, bronchitis and emphysema worsen due to higher pollution and longer, higher-pollen seasons.
11. **Expansion of Deserts:** desertification pushes deserts into previously productive rangelands.
12. **Spread of Infectious Diseases:** disease-carrying mosquitoes and ticks survive in new regions – malaria, dengue and Lyme disease spread.
13. **Bleaching of Coral Reefs (Acidification):** CO₂ absorbed by oceans forms carbonic acid, lowering pH; corals expel their algae, turn white and eventually die.

EFFECTS OF GLOBAL WARMING



The major effects of global warming.

5. What is Global Climate Change?

Climate Change

Climate change is a **long-term shift in global or regional climate patterns**. It includes changes in temperature, precipitation (rainfall, snowfall), wind patterns, sea levels, and the frequency and intensity of extreme weather events.

Why it Matters

Climate change is potentially **one of the greatest threats** to the environment, to biodiversity, and ultimately to the quality of human life.

While **global warming** (rising temperatures) is the primary driver of current climate change, **climate change** is the broader concept that encompasses all the resulting changes in Earth's climate systems.

6. Indicators of Climate Change

Eight indicators clearly demonstrate that the Earth's climate is changing.

6.1 Global Warming

- The increase in average temperature of atmosphere and ocean over the past 100 years – the **largest single indicator** of climate change.
- Some scientists believe the rate has **accelerated since the 1960s**.
- The 20th century was the warmest in 1,000 years, with records broken repeatedly in recent decades.

6.2 Changes in Polar and Glacial Ice

Large volumes of Earth's ice have been shrinking – one of the most visible, dramatic indicators. Impacts of melting ice:

- **(a) Rising Sea Levels:** melting glaciers and ice sheets add massive amounts of water to the oceans, threatening coastal regions worldwide.
- **(b) Habitat Loss for Animals:** **polar bears** depend on sea ice to hunt seals; melting ice makes food hard to find and their population is declining.
- **(c) Cultural & Lifestyle Disruption (Inuit People):** the **Inuit** of the Arctic traditionally travel and hunt **on sea ice**; thinning, disappearing ice makes this difficult and unpredictable, threatening their cultural survival.

6.3 Ocean Acidity

- Oceans have absorbed about **half of all CO₂** from fossil fuels over 200 years.
- Dissolved CO₂ forms **carbonic acid (H₂CO₃)**, lowering ocean pH – **ocean acidification**.

Effects: threatens corals, oysters, clams and sea urchins that build calcium-carbonate shells; affects reproduction of fish and plankton; since plankton is the base of the marine food web, this cascades through **entire marine ecosystems**. Coral reefs – which support about 25% of all marine species – are especially vulnerable.

6.4 Climate and Health

- **(a) Risk of Disease and Infection:** outbreaks increase in warmer periods – **malaria** mosquitoes thrive and expand; **plague** and other diseases increase.
- **(b) Increased Allergens:** more **dust** (droughts/dust storms), more **mould** (warm humid conditions), more **pollen** (longer seasons) – worsening allergies, asthma and respiratory conditions.
- **(c) Injury or Death from Extreme Weather:** heat waves, floods, hurricanes and storms directly cause injury and death. The **2003 European heat wave** alone claimed about **35,000 lives**.

6.5 Changing Wind Patterns

Changes in the distribution of heat across the surface have altered global wind patterns – **speed, frequency and direction** have fluctuated unpredictably for years. Unusual winds may have **accelerated Arctic ice melt** and also shift precipitation distribution.

6.6 Changing Precipitation Patterns

High temperatures increase evaporation; the warm, moisture-laden air rises, cools and produces rain or snow, altering global patterns – some regions get **more** (floods), others **less** (droughts).

Ontario, Canada

Since 1948, average annual temperatures across Ontario have risen by as much as **1.4°C**. Annual precipitation and the number of rainy days have increased; Northern Ontario has seen more snowfalls and greater snow quantity.

6.7 Storm Intensity and Frequency

Temperatures of tropical ocean waters (where tropical storms form) have risen by **0.33°C since 1981**, and the **strongest storms have become even stronger**. Warmer seas hold more energy, which converts into hurricane-speed winds – storm **intensity** (wind speed, rainfall) has risen even if the total number has not changed dramatically.

6.8 Changing Biomes

Biome

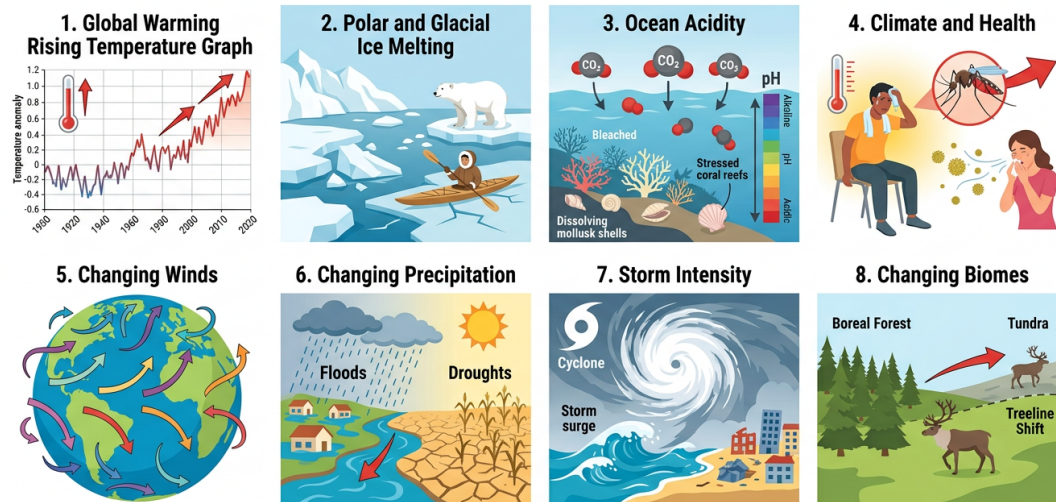
A **biome** is a large ecological community of plants and animals occupying a major habitat (e.g. tropical rainforest, grassland, tundra, desert).

As temperatures rise and rainfall shifts, biomes move **poleward and to higher elevations**; species that cannot migrate fast enough face extinction. **More than 1 million species** are threatened with extinction worldwide due to climate change.

7. Effect of Global Climate Change on Human Health

- **Heat Waves:** lead to heat stroke, dehydration and death – the **most common cause of weather-related deaths**. The 2003 European summer heat wave claimed about **35,000 lives**.
- **Allergies & Respiratory Health:** worsened by increased air pollution (ground-level ozone, particulate matter), higher and longer pollen seasons, and increased mould growth.
- **Spread of Diseases through Food, Water and Animals:** warmth and altered rainfall favour disease organisms and their vectors; flood contamination of water spreads water-borne diseases.

INDICATORS OF CLIMATE CHANGE



The eight indicators of climate change.

Global Statistics

WHO: climate-change-related factors (malaria, diarrhoea, malnutrition, floods) cause about **150,000 deaths worldwide annually**.

IPCC: about **250 million more Africans** are without potable (safe drinking) water due to climate-related stress.

Displacement of People: flooding linked to rising sea levels has displaced millions, creating **climate refugees** forced to migrate due to climate disasters.

8. Introduction to EIA

Environmental Impact Assessment (EIA)

“The study to **predict the effect of a proposed activity or project on the environment.**”

In simpler terms, EIA is a **systematic process** of evaluating the potential environmental consequences of a proposed project **before it is implemented**, so decision-makers can make informed choices.

8.1 What EIA Does

1. **Compares various alternatives** and identifies the one with the best combination of **economic and environmental costs and benefits**.
2. **Integrates environmental concerns** into development right at the **feasibility-report** stage, so concerns are addressed in the design and mitigation is built in from the start.
3. **Prevents future liabilities or expensive alterations** by catching problems early, when they are cheaper and easier to fix.

8.2 Purpose of EIA

- Ensure development decisions are made with **full knowledge** of likely environmental consequences.
- Provide decision-makers, developers and the public with information about environmental implications.
- Ensure **mitigation measures** are planned and implemented to minimise negative impacts.

9. Three Core Values of EIA

Core Value	Meaning
Integrity	The EIA process must be fair, objective, unbiased and balanced ; free of political or economic pressure; all information presented honestly.
Utility	The process must provide balanced, credible information that is useful and practical for decision-making, presented clearly.
Sustainability	The process must result in environmental safeguard , ensuring development does not compromise future generations – balancing development and protection.

10. Benefits of Environmental Assessment

1. **Protection of Environment:** prevents irreversible damage by identifying impacts before the project begins.
2. **Optimum Utilization of Resources:** ensures efficient, sustainable resource use.
3. **Saves Overall Time and Cost:** early problem detection avoids costly modifications and delays.
4. **Promotes Community Participation:** public hearings give local communities a voice.
5. **Informs Decision Makers:** provides comprehensive, scientifically sound information.
6. **Lays Base for Environmentally Sound Projects:** helps design projects that are both viable and responsible.

11. Evolution and History of EIA

- **Origin:** EIA began as a mandatory regulatory procedure in the early **1970s** with the **National Environmental Policy Act (NEPA) 1969** in the **United States** – the first law requiring federal agencies to consider environmental impacts before major decisions.
- **Spread:** EIA took off after the **mid-1980s**, when the **World Bank** adopted it as a requirement for the major projects it funded; borrower countries had to undertake EIA under the Bank's supervision.

- **Current Status:** EIA is now a formal, legally mandated process in **more than 100 countries**, each with its own framework.

EIA in India

Environmental Clearance from the central government is required for **32 categories** of development projects, including **Mining** (coal, iron ore, minerals), **Thermal power plants**, **River valley projects** (dams, irrigation, hydro-power) and **Infrastructure** (roads, highways, ports, harbours, airports).

12. EIA – Cycle and Procedure in India

The EIA process in India consists of **nine phases**:

The Nine Phases

1. Screening	Statement
2. Scoping and Consideration of Alternatives	6. Public Hearing
3. Baseline Data Collection	7. Environmental Management Plan
4. Impact Analysis	8. Decision Making
5. Mitigation and Environmental Impact	9. Monitoring the Clearance Conditions

12.1 Phase 1 – Screening

Screening is the **first stage**. It determines **whether** the project needs an EIA at all and, if yes, **what level** (comprehensive or rapid). **Screening criteria: Scale of Investment, Type of Development, and Location of Development.** Projects are classified into three categories:

Category	Description
A	Typically require a full, comprehensive EIA (by type, scale, location). May cause significant changes to land use, social, physical and biological environments. <i>Examples: large thermal power plants, large hydro dams, large mining projects, major ports.</i>
B	Differs from A only in scale – medium-sized projects, generally not in sensitive areas; mitigation is more easily prescribed as impacts are less complex.
C	Typically does NOT require an environmental assessment – small-scale, low-risk projects in non-sensitive locations.

12.2 Phase 2 – Scoping (and Alternatives)

Scoping **identifies key issues and significant impacts** to investigate, and **defines the boundary and time limit** of the study. It is done by a **consultant** in consultation with the **project proponent** and guidance from the regulatory **agency**.

- **Quantifiable impacts** are assessed by **Magnitude, Prevalence, Frequency and Duration**.
- **Non-quantifiable impacts** (aesthetic, recreational, cultural value) are also considered.
- **Significance** is usually determined through **socio-economic criteria**.
- Afterwards, areas of significant impact are identified, their **baseline status** is monitored, and **likely changes** from construction and operation are predicted.

12.3 Phase 3 – Baseline Data Collection

Baseline Data

Information about the **current state of the environment before the project begins** – a reference point against which future changes are measured.

Impact prediction “maps” the environmental consequences of the project and its alternatives. Impact can never be predicted with absolute certainty, but baseline data **reduces the degree of uncertainty**. Categories of baseline data:

- **Air:** changes in ambient and ground-level pollutant concentrations from **point, line and area sources**; effects on soils, materials, vegetation and human health.
- **Noise:** changes in ambient noise from equipment and vehicles; effects on fauna and human health.
- **Water:** availability to competing users; changes in water quality; sediment transport; ingress of saline water.
- **Land:** changes in land-use and drainage patterns; land quality (including waste disposal); shoreline/riverbank stability.
- **Biological:** deforestation and habitat shrinkage; impact on flora and fauna, on rare/endangered and endemic species, migratory paths, and breeding/nesting grounds.
- **Socio-economic:** impact on the local community (demographic changes), economic status (livelihoods) and human health.

12.4 Phase 4 – Impact Analysis

This phase involves the **detailed prediction and evaluation** of the identified significant impacts across all environmental components (air, water, land, biological, socio-economic) using appropriate scientific methods and models.

12.5 Phase 5 – Alternatives, Mitigation & EIA Report

12.5.1 Assessment of Alternatives

Identify possible alternatives and compare their environmental attributes – **alternative locations**, **alternative process technologies** (cleaner, less polluting), and the **“no project” alternative** (always considered). Alternatives are **ranked** and the best environmental option that also gives optimum economic benefit is selected.

12.5.2 Delineation of Mitigation Measures

A mitigation plan for the selected option is drawn up to **prevent/reduce** negative impacts or **compensate** for unavoidable ones, supplemented with an **Environmental Management Plan (EMP)**.

12.5.3 EIA Report

Gives the decision-maker information on different scenarios: “**with the project**”, “**without the project**”, and “**with project alternatives**”. **Uncertainties** must also be reflected honestly.

12.6 Phase 6 – Public Hearing

After the EIA report is prepared and made public, the **affected public must be informed and consulted**. The EIA summary is provided to:

- **Bona fide local residents** living in the affected area.
- **Local associations** (community/resident welfare organisations).
- **Environmental groups** active in the area.
- **Any person at the project site or sites of displacement**.

Purpose: transparency and democratic participation; allows communities to raise concerns; incorporates local knowledge. It is **mandatory in India** under the EIA Notification 2006.

12.7 Phase 7 – Environmental Management Plan (EMP)

EMP

A comprehensive plan detailing **how the environmental impacts identified in the EIA will be managed** throughout the project lifecycle.

It is critical for monitoring the **clearance conditions** and ensuring all **commitments** in the EIA report are implemented. Components:

1. **Delineation of Mitigation Measures:** prevention/control for each component (air, water, land, biology, socio-economy) plus rehabilitation and resettlement plans.
2. **Delineation of Monitoring Scheme:** what parameters are monitored, how often and by what methods, for compliance.
3. **Delineation of Implementation Plan:** scheduling and resource allocations (financial, human, technical).

12.8 Phase 8 – Decision Making

Through **consultation** between the **project proponent** (assisted by a consultant) and the **impact assessment authority** (e.g. the Ministry of Environment, Forest and Climate Change, assisted by an expert group). The decision is reached by evaluating the **EIA report** and **EMP**, reviewing public hearing comments and expert assessment.

Possible Outcomes

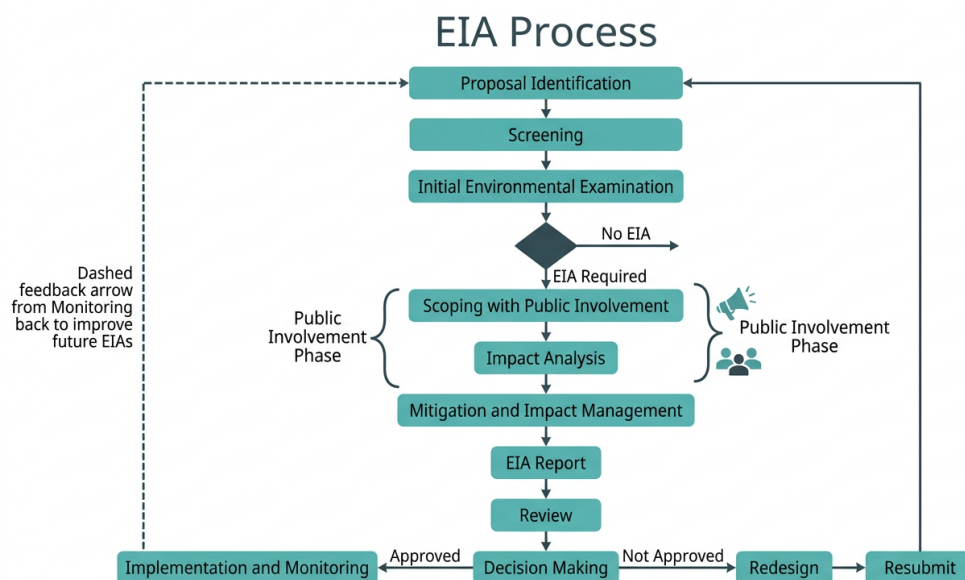
Approved (possibly with conditions) • **Not Approved** (rejected) • **Redesign Required**
 • **Resubmit** (more information/studies needed).

12.9 Phase 9 – Monitoring the Clearance Conditions

After clearance and start of the project, monitoring ensures commitments are kept.

- Must be done during **both** the **Construction phase** and the **Operation phase**.
- **Why:** to ensure commitments are complied with and that predictions were correct; if predictions were wrong (e.g. higher pollution), corrective action can be taken.
- **Corrective Actions:** if impacts exceed predicted levels or violate conditions – upgrade pollution control equipment, change procedures, or add mitigation.

13. EIA Process – Flowchart



The EIA process flowchart.

Note

Public involvement typically occurs at the Scoping / Impact-Analysis stage but can occur at any stage. Information from monitoring feeds back to improve future EIAs.

14. Components of EIA (EIA Report Contents)

14.1 Comprehensive EIA vs. Rapid EIA

Both vary only in the **time scale of data collected**:

Type	Description
Rapid EIA	Speedier appraisal using “ one season ” data (typically the most stressful season). Acceptable only if it does not compromise decision quality ; used when time is a constraint.
Comprehensive EIA	Uses data over all seasons (one full year or more) for a complete picture; required for projects with major, complex impacts.

Depending on the nature, location and scale of the project, the EIA report should contain all or some of the following components.

14.2 A. Air Environment

1. Determine the **impact zone** and develop a monitoring network (typically **7–10 km** from the site periphery).
2. Monitor existing **ambient air quality** (SO₂, NO_x, PM₁₀, PM_{2.5}, CO, etc.).
3. Monitor site-specific **meteorological data**: wind speed/direction, humidity, ambient temperature, environmental lapse rate.
4. Estimate **quantity and composition of air emissions** from the project.
5. Identify, quantify and evaluate **other existing emission sources** in the zone.
6. Predict changes in air quality via **air-quality dispersion modelling**.
7. Evaluate the **adequacy of proposed pollution-control devices**.
8. Delineate mitigation at **source, pathways and receptors**.

14.3 B. Noise Environment

1. Monitor the present (baseline) **noise levels** in the impact zone.
2. Predict future noise from machinery/equipment and vehicular movement.
3. Identify impacts on human health (hearing damage, stress, sleep disturbance) and wildlife (communication, breeding, migration).
4. Recommend mitigation: sound barriers/walls, noise-dampening equipment, operational time restrictions, vegetation buffers.

14.4 C. Water Environment

1. Study existing ground and surface water for **quality and quantity** within the impact zone.
2. Predict impact from pumping/extraction, drainage changes and contamination.
3. Quantify and characterise **wastewater** (volume and composition).
4. Evaluate the proposed **pollution-prevention and wastewater-treatment** system.
5. Predict impacts of **effluent discharge** on the receiving water body using hydrological and water-quality models.
6. Assess feasibility of **water recycling and reuse**.

14.5 D. Biological Environment

1. Survey **flora and fauna**, stating the season and duration of the survey.
2. Assess flora and fauna within the impact zone (species and abundance).
3. Assess potential damage to terrestrial and aquatic life from effluents and gaseous emissions.
4. Assess damage to terrestrial life from air pollution, land-use and landscape changes.
5. Assess damage to aquatic/marine life from physical disturbances and water- quality alterations.
6. Predict cumulative **biological stresses** in the impact zone.
7. Delineate mitigation: wildlife corridors, compensatory afforestation, restrictions in sensitive areas, protected-area buffers.

14.6 E. Land Environment

1. Study soil characteristics, existing land use, topography, landscape and drainage patterns in the impact zone.
2. Estimate impacts on land use, landscape, topography, drainage and hydrology.
3. Identify potential use of **treated effluent in land application** (e.g. irrigation) and its impacts.
4. Estimate and characterise **solid wastes** and management options (minimisation, environmentally compatible disposal – landfilling, incineration, recycling).

14.7 F. Socio-economic and Health Environment

1. Collect **demographic and socio-economic data** (population, employment, income, education, infrastructure).
2. Collect **epidemiological data** – endemic diseases and morbidity rates in the impact zone.
3. Project anticipated changes in socio-economic and health status due to the project.
4. Delineate measures to minimise adverse impacts (healthcare, livelihood restoration, community development).
5. Assess impact on historical, cultural and archaeological sites.
6. Assess economic benefits (employment, revenue, infrastructure, living standards).
7. Assess rehabilitation requirements, with emphasis on **scheduled areas**.

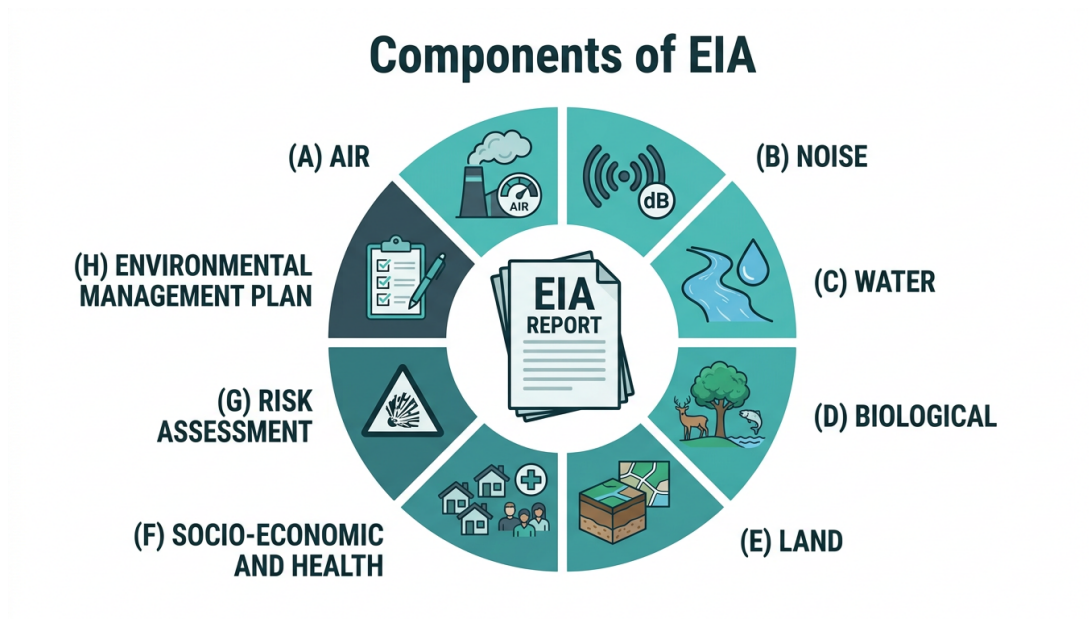
14.8 G. Risk Assessment

1. **Hazard Identification:** hazard indices, inventory analysis, dam-breaking probability, natural-hazard probability.
2. **Maximum Credible Accidents (MCA) Analysis:** worst-case plausible accident scenarios and their impact.
3. **Consequence Analysis:** fire, explosion, hazardous releases, dam breaks.

4. **Hazard and Operability (HAZOP) Studies:** systematic examination of operational processes for hazards.
5. **Assessment of risks** – ranking by probability and severity.
6. **Disaster Management Plan (DMP):** on-site (project personnel and infrastructure) and off-site (police, fire, administration, hospitals).

14.9 H. Environmental Management Plan (EMP)

(As detailed under Phase 7.) Includes delineation of **mitigation measures** (prevention/control plus rehabilitation and resettlement), a **monitoring scheme** for clearance compliance, and an **implementation plan** (scheduling and resource allocations).



The eight components of an EIA report.

15. EIA – Benefits and Flaws

15.1 Benefits of EIA

Benefit	Explanation
Systematic method of impact assessment	A structured, scientific approach ensuring nothing is overlooked.
Cost/benefit trade-off of alternatives	Compares options to find the best balance of economic benefit and environmental cost.
Facilitates public participation	Public hearings give communities a voice.
Effective coordination mechanism	Brings together departments, agencies and stakeholders.
Environmental integration	Integrates environmental concerns into planning from the start.
Negotiations	A platform for negotiating between developers and communities.
Feedback	Monitoring information improves future projects.
Top-level decision making	Informs high-level government approval decisions.
Balance of development and environment	The primary tool for environmentally responsible development.

15.2 Flaws of EIA

Flaw	Explanation
Time consuming	The process can take months or years, delaying urgent projects.
Costly	Requires significant funds for surveys, studies, consultations and reports.
Little public power in implementation	Communities often have little say in the actual decision or implementation.
Unreliable data (developing countries)	Depends on good baseline data, often lacking due to poor monitoring infrastructure.
Too focused on scientific analysis	May over-emphasise quantitative data at the expense of cultural/community concerns.
Poor post-clearance monitoring	Compliance monitoring after clearance is seldom carried out adequately.

15.

Summary Table – Module 5: Global Warming & EIA

Topic	Key Points
Global Warming Concept	Rising temperatures due to the enhanced greenhouse effect; GHGs trap heat; human activity increased GHG concentration.
Indicators of GW	Air surface temp, humidity, glaciers, snow cover, land temp, sea surface temp, sea ice, sea level, ocean heat, ocean temp.
Factors of GW	CO ₂ (power plants, vehicles), CH ₄ (rice, livestock), N ₂ O (fertilizers, industry), deforestation.
Effects of GW	Heat waves, changing rainfall, sea-level rise, storms, drought, food shortage, disease spread, coral bleaching.
Climate Change Indicators	Global warming, polar-ice changes, ocean acidity, climate–health link, wind patterns, precipitation, storm intensity, biome changes.
Health Impacts	Heat strokes, allergies, disease spread; WHO ~150,000 deaths/year; IPCC ~250 million without water.
EIA Definition	Study to predict the effect of a proposed project on the environment.
Core Values of EIA	Integrity, Utility, Sustainability.
EIA Benefits	Environment protection, resource optimization, cost saving, public participation, informed decisions.
History of EIA	Started USA 1970 (NEPA 1969); World Bank 1980s; now 100+ countries; India: 32 project categories.
EIA Phases	Screening → Scoping → Baseline Data → Impact Analysis → Mitigation/Report → Public Hearing → EMP → Decision → Monitoring.
Project Categories	A (large, full EIA), B (medium, simplified EIA), C (small, no EIA needed).
EIA Components	Air, Noise, Water, Biological, Land, Socio-economic, Risk Assessment, EMP.
EIA Flaws	Time consuming, costly, little public implementation power, data unavailability, poor post-clearance monitoring.

Module Complete

These are comprehensive, detailed study notes for **Module 5: Global Warming and EIA** (BCV755B – Conservation of Natural Resources), covering every topic of the module without skipping anything.